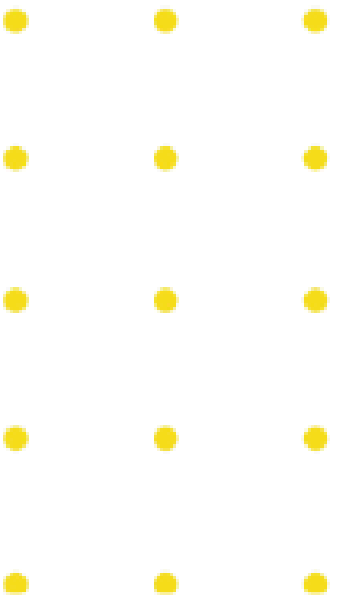


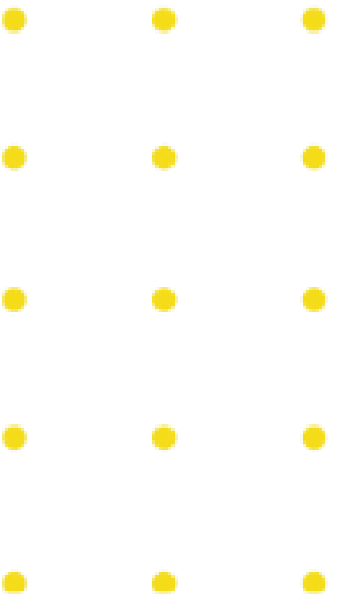
# **Module 3:**

# **Introduction to**

# **Network Security**



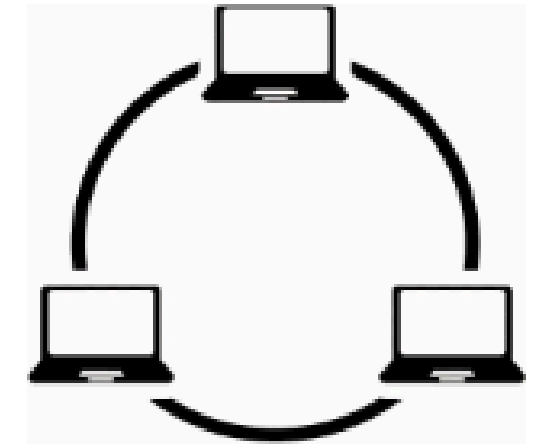
# Topic 1: Introduction to Network Security



# Network Fundamentals

## Types of Networks:

- **LAN:**
  - Also known as Local Area Networks, is a group of computers connected to each other locally or at one location.
- **WAN:**
  - Also known as Wide Area Networks, is a group of multiple LANs that has no limitations to city or country or else.
- **VPN:**
  - Also known as Virtual Private Networks, is a private network created over a public network and it creates an encrypted tunnel between network devices.



# Network Fundamentals

- **Important Network Protocols:**

- **TCP**
- **UDP**
- **ICMP**
- **ARP**
- **HTTP**
- **SSH**

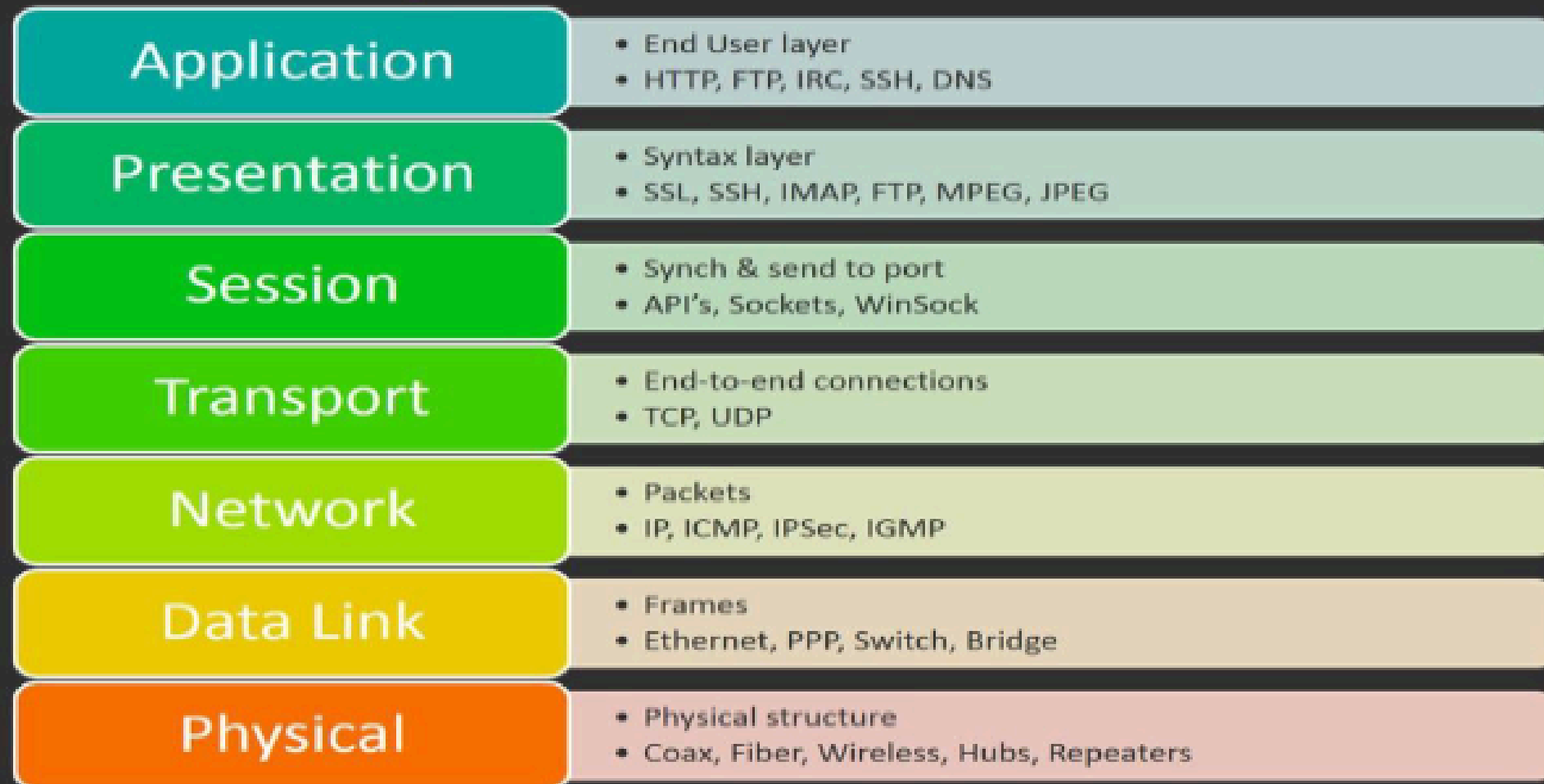
# Common Ports and Services

| Port Number | Transport Protocol | Service                     |
|-------------|--------------------|-----------------------------|
| 80          | TCP                | HTTP                        |
| 443         | TCP                | HTTPS                       |
| 20,21       | TCP                | FTP: File Transfer Protocol |
| 53          | UDP                | Domain Name Server          |
| 25          | TCP                | SMTP: Simple Mail Protocol  |
| 23          | TCP                | Telnet                      |



# Network Models

## 7 Layers of the OSI Model



OSI Model

TCP/IP Model